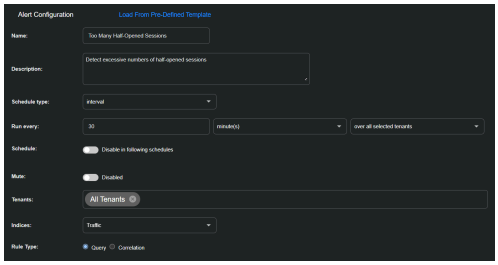
Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

ATH Example: Half-Opened Sessions

Create an Automated Threat Hunting (ATH) playbook to detect devices that formed unusually high numbers of half-opened sessions in a short period of time. A surge in half-opened sessions over a limited timeframe might indicate unstable communication behavior, misconfigured services, or the early stages of malicious activity. This example shows how to query traffic records for these patterns and configure the playbook to trigger when thresholds are exceeded.

Configure the Alert

Set up the alert to run at regular intervals so that the playbook can consistently identify devices exhibiting suspicious connection behavior. By querying for half-opened sessions every 30 minutes, you ensure timely detection of anomalies as they emerge in the network traffic.



To configure the alert:

1. Navigate to **Automation | Automation** and select the **Create** button to add a playbook.
2. Enter the following in the **Alert Configuration** section:

   Name: **Too Many Half-Opened Sessions**

   Description: **Detect excessive numbers of half-opened sessions**

   Schedule type: **interval**

   Run every: **30 minute(s) over all selected tenants**

   Schedule: Toggle off (default setting)

   Mute: Toggle off (default setting)

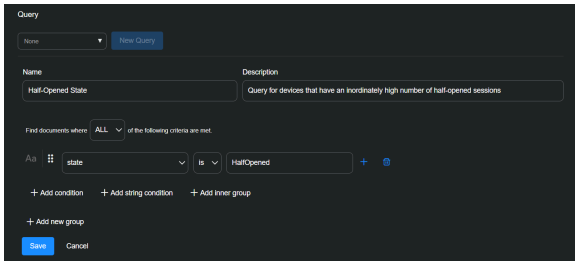
   Tenants: **All Tenants**

   Indices: **Traffic**

Rule Type: **Query**

Build a Query

Construct a query that identifies traffic in the half-opened state. This state occurs when a connection attempt is initiated but not completed—behavior that might suggest misconfigurations, network instability, or evasion tactics. To keep the query results manageable and actionable, limit the number of source IP addresses returned to the top 1,000 by frequency.



1. Select **New Query**.
2. Enter the following and then select **Save**:

   Query Name: **Half-Opened State**

   Description: **Query for devices that have an inordinately high number of half-opened sessions**

   Find documents where **ALL** of the following criteria are met.

   Select **+ Add condition** and enter **state**, **is**, **HalfOpened**.
3. Toggle **Calculations** on.
4. Select **+** to add a calculation.
5. Enter the following:

   Name: **Top List**

   Calculation type: **Top**

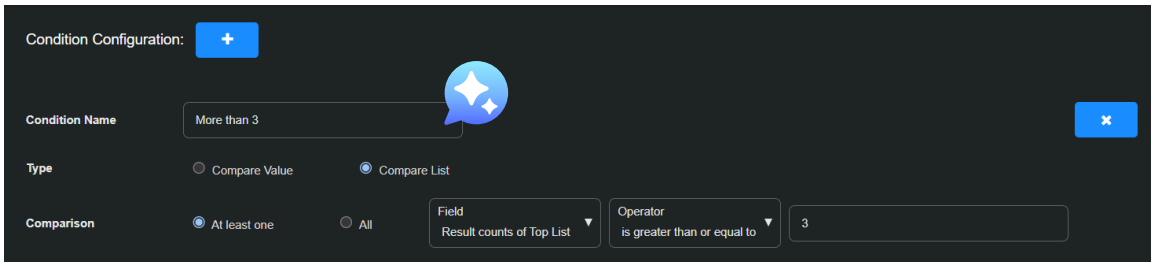
   Field to calculate on: **srcip**

   Calculation By: **count**

   Size: **1000**

Configure a Condition

Define a threshold that indicates when the observed behavior merits attention. In this example, if the number of unique source IP addresses generating half-opened sessions exceeds three, the condition triggers. This allows actions to be taken only when the behavior appears persistent or widespread enough to justify investigation.



Condition Name: **More than 3**

Type: **Compare List**

Comparison: **At least one**, **Result counts of Top List**, **is greater than or equal to**, **3**

Configure Actions

You can configure any number of actions to take place if your playbook gets the specified number of hits. Remember to set the **Trigger on condition** to the condition you created (which is automatically selected). In this example, it's **More than 3**.

See Configure Actions for details on configuring each type of action.

Save & Run the Playbook

Select **Submit** to save and immediately run the playbook.

The new playbook appears in the Automated Threat Hunting Playbooks list.

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